

HyPER Oil-Pyrolysis to Hydrogen Fuel: Integrated Feasibility Study & Project Plan

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An essay submitted to the module of Engineering Design and Project Management in partial fulfilment of the requirement of the degree of

MSc Advanced Mechanical Engineering

Faculty of Engineering and Applied Sciences

Tutors: Dr Gill Drew, Dr Adriana Encinas-Oropesa

Academic Year 2025-2026

Project Charter: HyFuel-Pyro

Project manager	Isaac Summers 492991		
Project Internal Sponsor	Cranfield University - HyPER Project		
Project Start Date	5th March 2026	Target end Date	28th July 2026

Project Summary: (…what are you going to do)

Design and plan a feasibility study for an integrated energy system that uses oil-pyrolysis with the HyPER pilot plant and produces low-carbon hydrogen for use as a fuel. The project will define the technical concept, supply and offtake arrangements, project management approach and risk framework required to progress from concept selection to a demonstration project at Cranfield. The pathway will focus on hydrogen as a fuel for decarbonising a specific end use sector (e.g. industrial heat or transport) while managing carbon emissions from the integrated system.

Project Justification / business case (…why)

Hard to abate sectors that rely on fossil fuels for heat and power, have increasing pressure to cut emissions but often lack access to affordable low-carbon at scale. Integrating oil-pyrolysis with the HyPER SE-SMR technology offers a route to produce low carbon hydrogen fuel with high CO_2 capture rates and competitive levelised cost compared with conventional SMR-CSS or ATR-CSS. The project will test whether this pathway can supply hydrogen fuel with a significantly reduced carbon footprint, giving value through avoided emissions costs, improved compliance with schemes such as UK ETS and potential application in hydrogen and carbon markets.

Project Constraints (… within what limits)

Time	Study must be completed within a 24 month window to align with HyPER pilot availability and funding cycles.
Scope	Limited to a single integrated pathway: oil-pyrolysis feedstock linked to the HyPER pilot and one primary hydrogen-fuel application at or near Cranfield. Excludes full FEED and construction; focusses on concept design, techno economic assessment, risk and project management framework for a future demonstration project.
Quality	Hydrogen product must meet fuel grade purity and safety requirements for the chosen end use. Feasibility outputs must be robust enough to support external stakeholder review (industry partners, funders, regulators), using recognised engineering and PM standards.
Cost	Feasibility study constrained by a fixed budget envelope (to be agreed with sponsors), prioritising analysis that significantly reduces technical or commercial uncertainty. Any proposed future CAPEX/OPEX must give a credible route to competitive LCOH compared with current fuel options and alternative low carbon hydrogen technologies.

Outcomes (…to deliver what specifics)	Success Measure (baseline and target if applicable)
Defined integrated pathway concept	Approved process and system architecture for the pathway, including block flow diagrams and key design parameters.
Techno-economic and environmental performance case	Quantified ranges for hydrogen production rate, CO_2 capture rate and symbolic LCOH and a qualitive assessment of lifecycle emissions.
Project management and delivery plan	Completed project plan (WBS, Gantt, milestones) which fits internal PM standards.
Risk and stakeholder framework	Initial risk register with owners and mitigations and a mapped stakeholder set with clear value propositions for each key party.

Stakeholders Name (…who for)	Stakeholder Function	Stakeholder Goal
Cranfield University	Technology owner	Demonstrate flexibility of HyPER and secure future partnerships
Industrial hydrogen fuel user	Anchor hydrogen offtaker	Access reliable, competitively priced low carbon hydrogen fuel to reduce emissions and costs
Oil-pyrolysis feedstock provider	Feedstock supplier	Secure long term offtake contracts

Key Starting Risks to success

Risk ID#	Risk	Mitigation
1	Insufficient demand from the chosen hydrogen end-user	Early market engagement with potential offtakers
2	Cost and funding constraints	Seek co funding or in-kind industrial support
3	Feedstock supply or quality issues	Engage multiple waste oil/pyrolysis suppliers early
4	Regulatory or permitting delays for H_2 pipeline	Early engagement with HSE local authorities and pipeline regulators

Project Design - HyFuel-Pyro Pathway (Oil-Pyrolysis H_2 as Fuel)

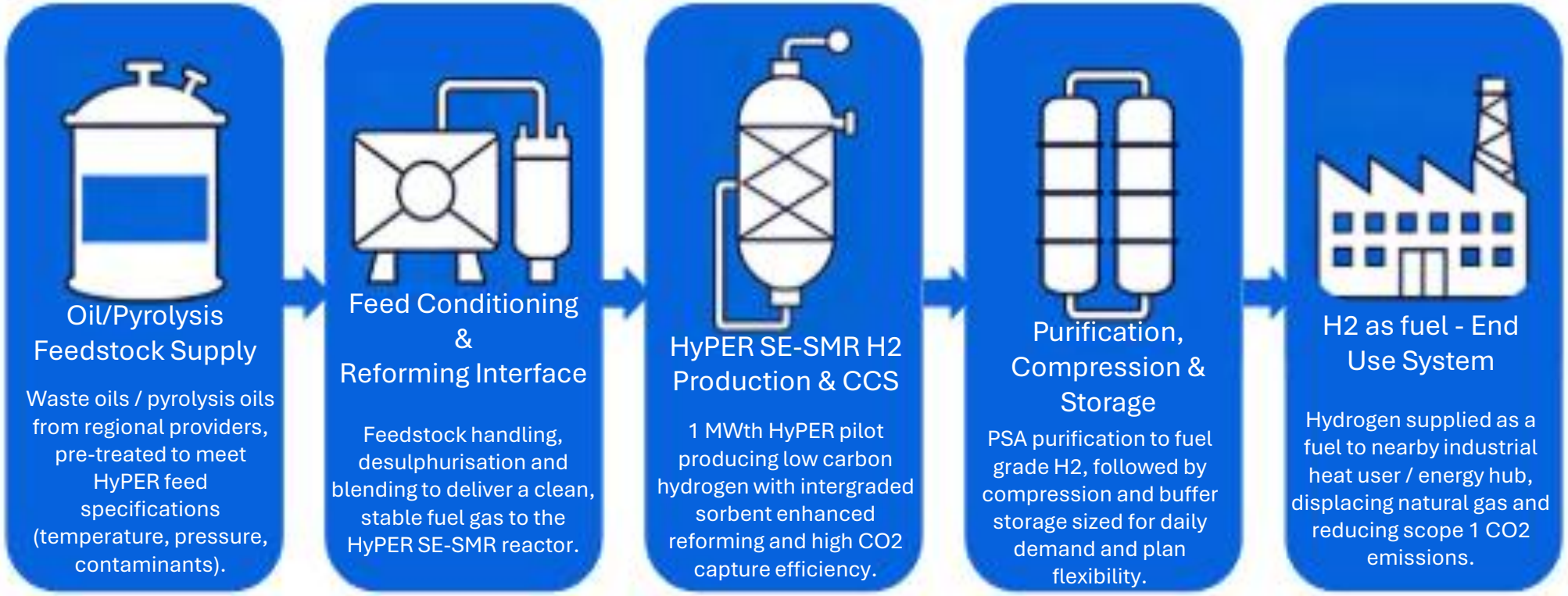
Design Characteristics

Integrated Pathway:
Couple regional oil-pyrolysis feedstock with HyPER pilot at Cranfield

Continuous operation:
~8,000 h/yr with planned shutdowns, aligned with industrial H2 plant practice

Full plant scope:
Feed handling, reaction, separation, utilities and delivery - not just reactor

Safety and Ethics by design:
Hydrogen hazards, operator needs, regulatory requirements embedded from project start



Transport and Interfaces

Feedstock Transport:
Road tanker delivery (ADR-compliant) from regional suppliers to site.
Hydrogen Delivery:
Short dedicated pipeline to industrial heat user (food processing/light manufacturing).

On site integration:
High pressure storage → compressors → pipeline injection
Utilities interfaces: electricity ↔ compressors/control; steam ↔ process units; SCADA ↔ safety systems

Key Resources & Supporting Systems

Utilities:
Steam, electricity, cooling water, instrument air (sized for operations and compression)

Instrumentation and control:
Central DCS for monitoring temperature, pressure, flow, gas composition, automated start-up/shutdown

Safety systems:
H2/CO/hydrocarbon detection, fire/explosion protection, emergency shutdown, relief and flare systems.

Environmental monitoring:
CO2 and NOx emissions tracking for permit compliance and certification.

Human resources:
Operations team, maintenance technicians, HSE specialists, project management.

Justification Slide 1

H2 Purification, Storage & Delivery Justification

Why PSA purification + compression:

Fuel grade specification: PSA purifies HyPER output (~94% H2 from reformer) to >99.9% purity required for industrial burners and fuel cells [13].

Dual-pressure capability: Compression enables both medium pressure pipeline delivery (20-30 bar) and high pressure refuelling (350-700 bar) from a single source [18].

Operational flexibility: Buffer storage sized for daily demand variations and plant start-up/shutdown sequences [18].

Why dedicated pipeline to industrial user:

- **Lowest levelised cost:** £0.3-0.5/kg H2 for pipeline vs £1.0-1.5/kg for tube trailers over 5-10 km sustained delivery [App C]
- **Operational simplicity:** Direct connection eliminates daily logistics, driver scheduling, trailer availability constraints [App C].
- **Scalability:** Same pipeline infrastructure supports future scale up to 50+ MWth commercial plant with pressure boosting [App C].
- **Regulatory alignment:** UK actively enabling H2 pipeline rollout and transport frameworks for industrial clusters [17]
- **Demonstration value:** Proves low cost H2 delivery model for UK industrial decarbonisation pathways [App C].

Why industrial heat end use:

- **Direct scope-1 decarbonisation:** Substitutes natural gas in existing burners/boilers (H2-ready or retrofitted), resulting in immediate emissions reductions [App B]
- **Steady demand profile:** Continuous baseload heat demand (food processing, light manufacturing) enables optimal HyPER operation at design efficiency [App B]
- **Economic driver:** Industrial users face rising carbon costs (UK ETS £49/t CO2 and climbing), H2 fuel provides compliance pathway [App B].
- **Offtake agreement:** Structured long term H2 supply contract de-risks pilot economics and validates commercial business model [App B]
- **Secondary Cranfield refuelling access:** Provides demonstration/outreach value and future mobility flexibly without compromising primary industrial supply.

Alternative rejected: Transport fuel as primary end use (intermittent refuelling demand unsuitable for continuous pilot operation).

Feedstock Selection & Transport Justification

Why oil/pyrolysis feedstock:

- **Circular economy alignment:** Valorises regional waste streams (East of England has sufficient volume to fulfil estimated 2-3 tonnes oil/pyrolysis feedstock per day requirement, with over 200k+ t/yr in unused cooking oil (UCO) alone nationally [App A], replaces conventional fossil feedstock.
- **Carbon intensity benefit:** Waste feedstock reduces lifecycle carbon footprint vs fossil duel natural gas while maintaining SE-SMR process compatibility [13].
- **Technology readiness:** HyPER/CHG process proven for liquid hydrocarbon feedstocks with standard pre-treatment (desulphurusation, volatiles control) [13].
- **Cost-effectiveness:** Waste oils competitively priced vs other sustainable alternatives such as biomethane. [App A]
- **Scalability:** National waste oil volumes (>200k t/yr) support future 50-300 MWth commercial deployment in industrial in industrial clusters [App A].

Why road tanker delivery:

- **Infrastructure reality:** No existing waste oil pipeline network, so ADR compliant tanker delivery is used and is established practice for Class 3/9 liquids [App C].
- **Operational flexibility:** Schedulable delivery (~50 trips/yr at 20 t/load) aligns with batch storage and pilot operational patterns [App C].
- **Short haul economics:** Road transport cost effective for <50km distances and pilot scale volumes (~650-1,000t/yr) [App C].
- **Negligible carbon:** Feedstock transport emissions ~2.7 t CO2e/yr (0.1% of plant's carbon benefit) [App C]

Alternative rejected: Biomethane (limited regional availability, competing demand for transport/heat, higher feedstock cost due to limited grid injection capacity) [App A]

HyPER Process & Scale Justification

Why 1.5 MWth pilot scale:

- **TRL progression:** Appropriate for Phase 2 TRL 4-6 demonstration, 20x scale up from GTI's existing 71 kWth US facility [13]
- **Risk mitigation:** Pilot scale validates elevated pressure operation (4-24 bara) and continuous sorbent cycling before commercial deployment [13]
- **User matched capacity**: ~700 kg H2/day sized for single medium scale industrial heat user (5-10 MW thermal demand) [App B], which allows for steady continuous operation (~8,000 h/yr).

Why HyPER/SE-SMR technology:

- **Superior techno-economics:** ~56% lower CAPEX (£105M vs £240M for 200MWth plant) and ~20% lower LCOH (£170 vs £212/kNm$^{3)}$ vs SMR+CCS counterpart [13].
- **Higher CO2 capture:** Up to 98% capture via integrated sorbent enhanced reforming and calciner vs 90% for conventional SMR+CCS [13].
- Low carbon footprint: HyPER process achieves ~40% of SMR+CCS carbon footprint (38 vs 90+ kg CO_2/kNm^3 H_2) which allows for near zero emissions pathway with further optimisation. [13]
- Cranfield capability: Leverages existing HyPER pilot infrastructure, TRL 2-6 expertise and hydrogen research facilities at site.

Process design approach:

- **Full-plant integration:** Feed handling → Reaction → Separation → Utilities → Delivery; reflects industrial H2 plant practice and design integration mapping [18].
- **Continuous operation:** ~8,000 h/yr with planned shutdowns aligns with large scale H2 plant operational practice [13].

Justification Slide 2

Strategic Fit & Scale Up Pathway

Strategic location & clusters:

- Cranfield sits within reach of regional waste-oil/pyrolysis supply and multiple potential industrial heat users (5-50km), making it realistic test bed for future cluster projects rather than an isolated demo [App A, App B].
- Existing HyPER pilot and H2 refueling infrastructure mean the site can host TRL 4-6 demonstrations without greenfield build out.

Infrastructure & regulation:

- The pathway is aligned with emerging UK hydrogen infrastructure plans (core hydrogen network, cluster pipelines) and policy statements allowing for hydrogen pipeline rollout and industrial decarbonisation infrastructure [17].
- Demonstrating safe operation of waste to H2 + pipeline + industrial user integration at Cranfield directly suppports future HyNet/cluster style deployments [App C, 1]

Scale up logic:

- Technical and commercial learning from a 1.5 MWth/~700 kg H2/day pilot directly informs the roadmap of 10MWth → 50MWth → 300MWth HyPER units set out in the Phase 1 study, designed for UK industrial clusters by ~2030 [13].
- The design keeps options open for future feedstock switching (biomethane) and additional end uses (mobility, power) without redesigning the core H2 production train. [App A]

Supporting Systems & Resources Justification

Utilities sizing rational:

- Steam, electricity, cooling water, instrument air scaled per industrial H2 plant best practice to support SE-SMR reactor, calciner thermal loads and compression energy requirements.
- Ensures reliable continuous operation and process stability.

Safety systems design:

- **Hydrogen specific hazards addressed:** H2/CO/hydrocarbon gas detection, fire/explosion protection (flame detection, water deluge), emergency shutdown (ESD) systems, relief and flare.
- **Standards compliance:** Design aligns with UK HSE guidance for hydrogen facilities, ATEX directives for explosive atmospheres and PED pressure equipment [19].
- **User centered approach:** Operator needs, control room ergonomics and maintenance access embedded from outset are considered.

Instrumentation & control:

- Central DCS monitors temperature, pressure, flow and gas composition across all process units.
- Automated startup/shutdown sequences reduce operator burden and improve safety.
- SCADA integration enables real-time monitoring and remote diagnostics.

Environmental monitoring:

- CO2 and NOx emissions tracking supports permit compliance and future Low Carbon Hydrogen certification (30 g CO2e/MJ lifecycle emissions) [App B].
- Enables verification of 98% CO2 capture performance and plant carbon intensity claims.

Human resources:

- Operations team (HyPER + supporting units), maintenance technicians, HSE specialists, project management for stakeholder coordination.
- Skilled workforce requirement supports regional clean energy employment and knowledge [App B].

Design principles applied:

- **Full system integration:** Demonstrates complete energy pathway from waste feedstock through H2 production to end user decarbonisation.
- **Industrial practice:** Clear separation of feed handling, reaction, gas separation, utilities and delivery reflects commercial H2 plant structure.
- **Safety & ethics by design:** Hazards, operator needs, regulatory requirements considered from project start.
- **Future proofing:** Pilot operation generates engineering data to de risk scale up to 50-300 MWth commercial plants for UK industrial clusters [13].

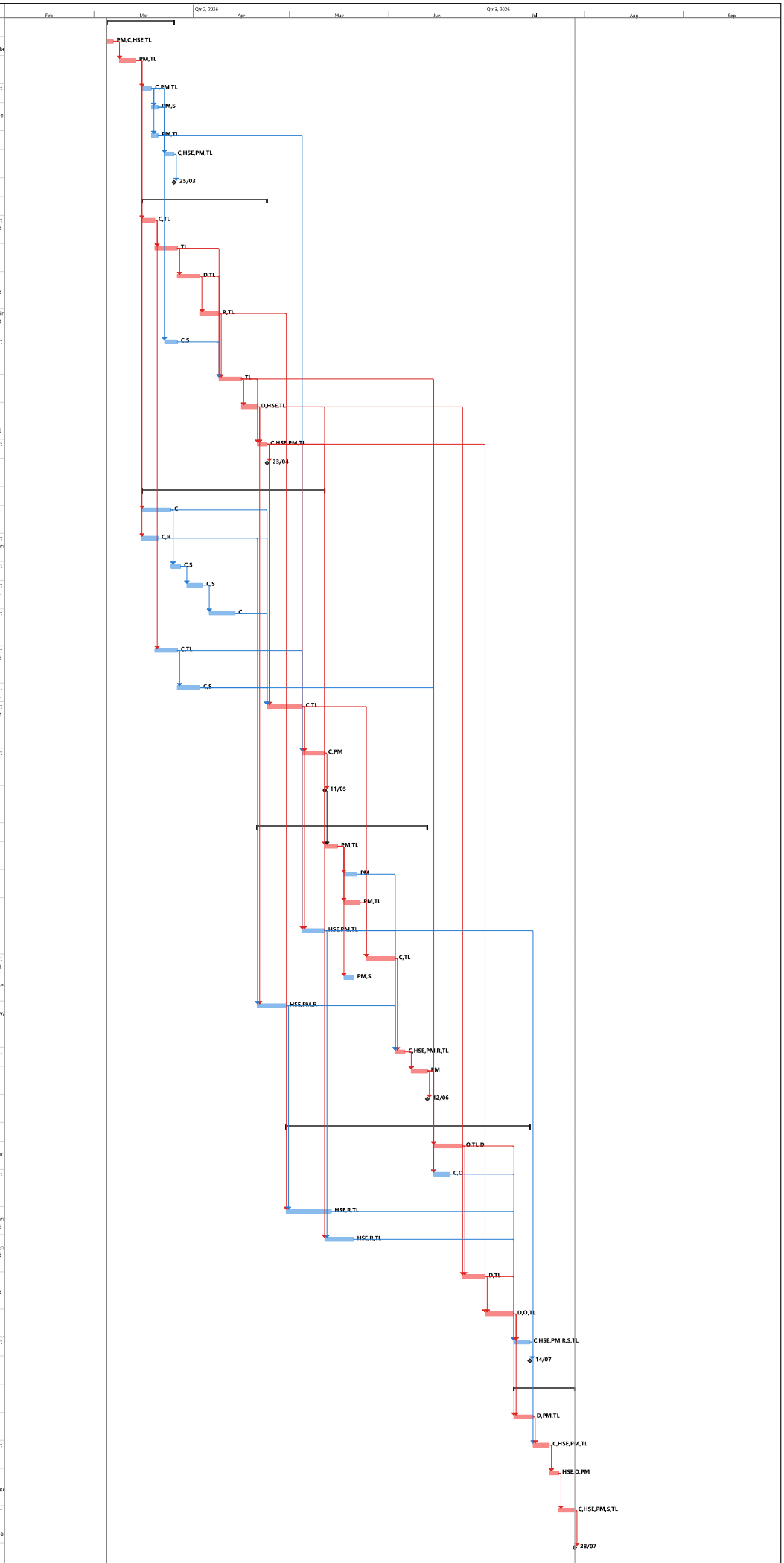
Economic and Financial Case [App D]

- **Capacity and production:** Pilot plant sized at ~700 kgH2/day, operating ·at ~8,000 h/yr (88% utilisation) → ~225-230 tonnes H_2/yr supplied to industrial heat user.
- **H_2 revenue potential:** Order of £0.9M/yr at central estimate of £4.00/kg (range of £3.50-£5.00/kg for delivered low carbon H_2 in UK industrial market). [14]
- **Carbon displacement value:** ~2,100 tonnes CO_2/yr avoided (vs natural gas heat + conventional SMR emissions), worth ~£103k/yr at UK ETS price of £49/t CO_2. [8]
    - **Combined annual benefit:** order of £1.0M/yr
- **Techno-economic competitiveness:** HyPER/GHC technology achieves up to 98% CO_2 capture with ~20% lower LCOH and >50% CAPEX reduction vs SMR+CCS at scale which supports long term cost viability and commercial scalability. [13]
- **Risk mitigated approach:** Pilot scale demonstration (700kg/day) de risks future commercial deployment at 50-300 MWth scale for UK industrial clusters.

Social and Stakeholder Value

- **Job creation:** Skilled operations, maintenance and HSE roles at regional level, supports local employment in clean energy transition.
- **Industrial user benefit:** Provides affordable access to low carbon H2 fuel, improving competitiveness under tightening UK ETS regulations (carbon price £49/t CO2 and rising)
    - Enables scope-1 emissions reduction for food processing, light manufacturing and other industrial heat users.
- **University mission alignment:** Demonstrates Cranfield's hydrogen leadership and TRL 2-6 capability and leverages existing HyPER pilot infrastructure and expertise.
- **Community and safety:** Safe, clean energy demonstration with minimal visual/noise/air quality impact vs alternatives, rigorous HSE management and stakeholder engagement.
- **Skills and knowledge transfer:** Generates operational data and engineering insights for UK industrial decarbonization, supporting workforce upskilling in hydrogen technologies.

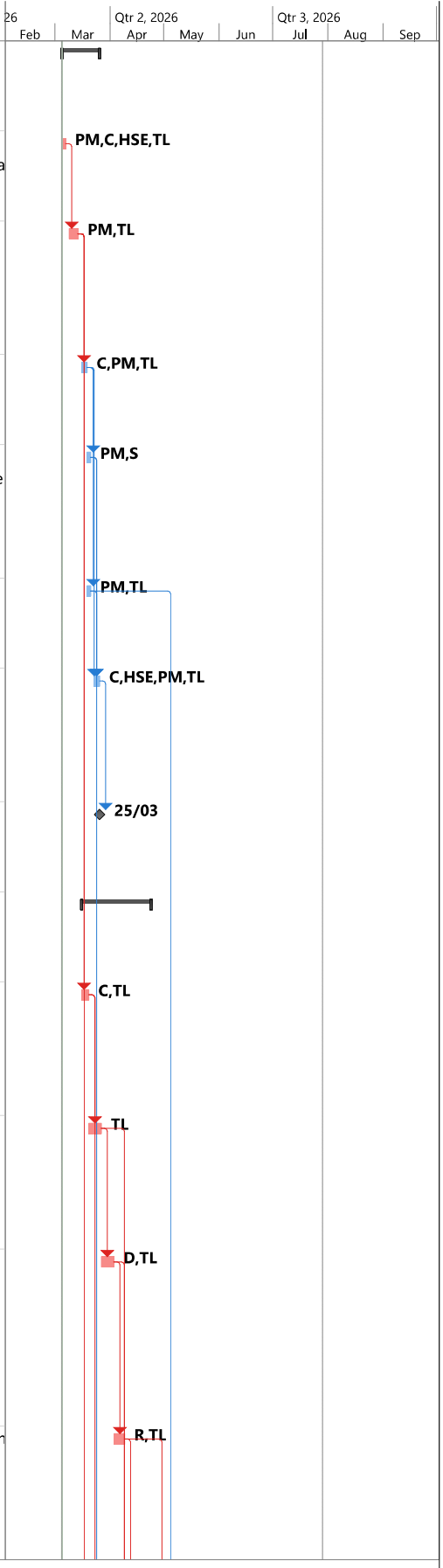
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1	1		**Project preperation & definition**	**15 days**	**Thu 05/03/26**	**Wed 25/03/26**		
2	1.1		Project initiation meeting	2 days	Thu 05/03/26	Fri 06/03/26		Project Manager,Commercia
3	1.2		Review HyPER/Oil-Pyro documentation	5 days	Mon 09/03/26	Fri 13/03/26	2	Project Manager,Technical Lead
4	1.3		Define project objectives &	3 days	Mon 16/03/26	Wed 18/03/26	3	Commercial/Market Lead,Project
5	1.4		Identify key stakeholders and roles	2 days	Thu 19/03/26	Fri 20/03/26	4	Project Manager,Stakeholde & Comms Lead
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7	1.6		Finialise Project Charter and Internal sign off	3 days	Mon 23/03/26	Wed 25/03/26	5,4,6	Commercial/Market Lead,HSE Lead,Project
8	1.7		Milestone 1: Charter approved	0 days	Wed 25/03/26	Wed 25/03/26	7	
9	2		**Concept & Technical design**	**29 days**	**Mon 16/03/26**	**Thu 23/04/26**		
10	2.1		Define oil/pyrolysis feedstock	4 days	Mon 16/03/26	Thu 19/03/26	3	Commercial/Market Lead,Technical Lead
11	2.2		Define HyPER-feed interface and	5 days	Fri 20/03/26	Thu 26/03/26	10	Technical Lead
12	2.3		Define H2 purification, compression & storage scheme	5 days	Fri 27/03/26	Thu 02/04/26	11	Data & Instrumentation Lead,Technical Lead
13	2.4		Define H2 delivery route (pipeline	4 days	Fri 03/04/26	Wed 08/04/26	12	Regulatory/Permittin Lead,Technical Lead
14	2.5		Define industrial heat end-user requirements (heat load,	4 days	Mon 23/03/26	Thu 26/03/26	5	Commercial/Market Lead,Stakeholder & Comms Lead
15	2.6		Develop integrated block flow	5 days	Thu 09/04/26	Wed 15/04/26	11,12,	Technical Lead
16	2.7		Identify supporting sytems (utilities, safety, I&C,	3 days	Thu 16/04/26	Mon 20/04/26	15	Data & Instrumentation Lead,HSE Lead,Technical Lead
17	2.8		Technical review of concept design	3 days	Tue 21/04/26	Thu 23/04/26	16,15	Commercial/Market Lead,HSE
18	2.9		Milestone 2: Concept design chosen	0 days	Thu 23/04/26	Thu 23/04/26	17	
19	3		**Market, stakeholder &**	**41 days**	**Mon 16/03/26**	**Mon 11/05/26**		
20	3.1		Market analysis for industrial H2 fuel in region	7 days	Mon 16/03/26	Tue 24/03/26	3	Commercial/Market Lead
21	3.2		Policy & regulatory review (UK ETS, H2	5 days	Mon 16/03/26	Fri 20/03/26	3	Commercial/Market Lead,Regulatory/Per Lead
22	3.3		Identify candidate	3 days	Wed 25/03/26	Fri 27/03/26	20	Commercial/Market Lead,Stakeholder &
23	3.4		Initial engagement with target industrial	5 days	Mon 30/03/26	Fri 03/04/26	22	Commercial/Market Lead,Stakeholder & Comms Lead
24	3.5		Develop outline H2 offtake concept (volumes, pricing, term)	6 days	Mon 06/04/26	Mon 13/04/26	23	Commercial/Market Lead
25	3.6		Feedstock supply mapping (waste oil/pyrolysi providers)	5 days	Fri 20/03/26	Thu 26/03/26	10	Commercial/Market Lead,Technical Lead
26	3.7		Initial feedstock supplier	5 days	Fri 27/03/26	Thu 02/04/26	25	Commercial/Market Lead,Stakeholder &
27	3.8		High level techno-economic assessment (H2 revenue, CO2 value, CAPEX/OPEX	7 days	Fri 24/04/26	Mon 04/05/26	26,24,	Commercial/Market Lead,Technical Lead
28	3.9		Draft commercial/partn model (roles, responsibilities)	5 days	Tue 05/05/26	Mon 11/05/26	27,25	Commercial/Market Lead,Project Manager
29	3.10		Milestone 3: Commercial & stakeholder concept agreed in principle	0 days	Mon 11/05/26	Mon 11/05/26	28	
30	4		**Detailed planning & approvals**	**39 days**	**Tue 21/04/26**	**Fri 12/06/26**		
31	4.1		Develop work breakdown structure (WBS)	4 days	Tue 12/05/26	Fri 15/05/26	29,17	Project Manager,Technical Lead
32	4.2		Develop detailed project schedule (Gantt)	4 days	Mon 18/05/26	Thu 21/05/26	31	Project Manager
33	4.3		Resource planning (people, equipment,	5 days	Mon 18/05/26	Fri 22/05/26	31	Project Manager,Technical Lead
34	4.4		Refine risk assessment & mitigation plan	5 days	Tue 05/05/26	Mon 11/05/26	27,17,	HSE Lead,Project Manager,Technical Lead
35	4.5		Cost estimate and budget	7 days	Mon 25/05/26	Tue 02/06/26	33,27	Commercial/Market Lead,Technical Lead
36	4.6		Define governance and reporting	3 days	Mon 18/05/26	Wed 20/05/26	31	Project Manager,Stakeholde & Comms Lead
37	4.7		Prepare regulatory and permitting strategy (safety, environment, pipeline)	7 days	Tue 21/04/26	Wed 29/04/26	21,16	HSE Lead,Project Manager,Regulatory, Lead
38	4.8		Internal review of plan and	3 days	Wed 03/06/26	Fri 05/06/26	37,35,	Commercial/Market Lead,HSE
39	4.9		Client/sponsor approval of project plan	5 days	Mon 08/06/26	Fri 12/06/26	38	Project Manager
40	4.10		Milestone 4: Project plan approved /	0 days	Fri 12/06/26	Fri 12/06/26	39	
41	5		**Implementation and pilot**	**54 days**	**Thu 30/04/26**	**Tue 14/07/26**		
42	5.1		Site integration planning (layout, tie-ins,	7 days	Mon 15/06/26	Tue 23/06/26	39,15	Operations Representitive,Techr Lead,Data &
43	5.2		Feedstock logistics plan (contracts, tanker schedules, storage)	5 days	Mon 15/06/26	Fri 19/06/26	39,26	Commercial/Market Lead,Operations Representitive
44	5.3		H2 pipeline pre FEED study (route,	10 days	Thu 30/04/26	Wed 13/05/26	37,13	HSE Lead,Regulatory/Per Lead,Technical Lead
45	5.4		HSE management plan (hazards, controls, emergency	7 days	Tue 12/05/26	Wed 20/05/26	34,16	HSE Lead,Regulatory/Per Lead,Technical Lead
46	5.5		Data and instrumentation plan (what to measure, when and how)	5 days	Wed 24/06/26	Tue 30/06/26	42,16	Data & Instrumentation Lead,Technical Lead
47	5.6		Draft pilot test plan (objectives, KPIs, run matrix)	7 days	Wed 01/07/26	Thu 09/07/26	46,17	Data & Instrumentation Lead,Operations
48	5.7		Implementation and readiness	3 days	Fri 10/07/26	Tue 14/07/26	47,45,	Commercial/Market Lead,HSE
49	5.8		Milestone 5: Ready for pilot implementation	0 days	Tue 14/07/26	Tue 14/07/26	48	
50	6		**Monitoring, evaluation and close out**	**13 days**	**Fri 10/07/26**	**Tue 28/07/26**		
51	6.1		Define monitoring and reporting	4 days	Fri 10/07/26	Wed 15/07/26	47,46	Data & Instrumentation Lead,Project
52	6.2		Define success criteria and evaluation plan	3 days	Thu 16/07/26	Mon 20/07/26	51,34	Commercial/Market Lead,HSE Lead,Project
53	6.3		Lessons learnt and continuous improvement process	3 days	Tue 21/07/26	Thu 23/07/26	52	HSE Lead,Operations Representitive,Projec Manager
54	6.4		Final project review and handover to implementation	3 days	Fri 24/07/26	Tue 28/07/26	53	Commercial/Market Lead,HSE Lead,Project Manager,Stakeholde & Comms
55	6.5		Milestone 6: Project definition and plannnig	0 days	Tue 28/07/26	Tue 28/07/26	54	



Project: Assign Gannt.mpp
Date: Thu 05/03/26

Task | Split | Milestone | Summary | Project Summary | Inactive Task | Inactive Milestone | Inactive Summary | Manual Task | Duration-only | Manual Summary Rollup | Manual Summary | Start-only | Finish-only | External Tasks | External Milestone | Deadline | Critical | Critical Split | Progress | Manual Progress

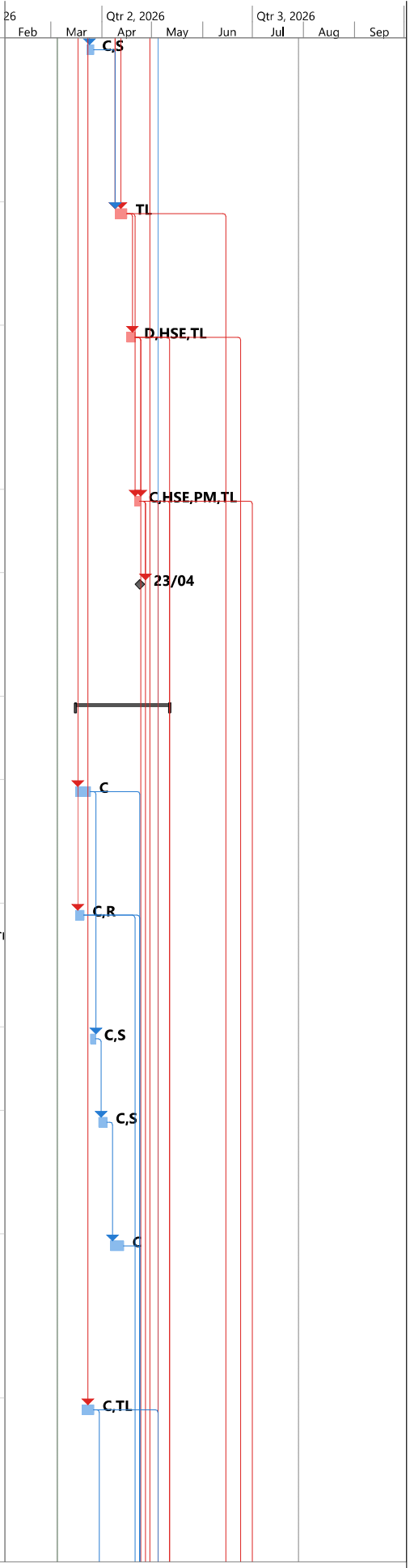
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Date: Thu 05/03/26

Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Critical	
Project Summary	Manual Summary	Critical Split	
Inactive Task	Start-only	Progress	
Inactive Milestone	Finish-only	Manual Progress	

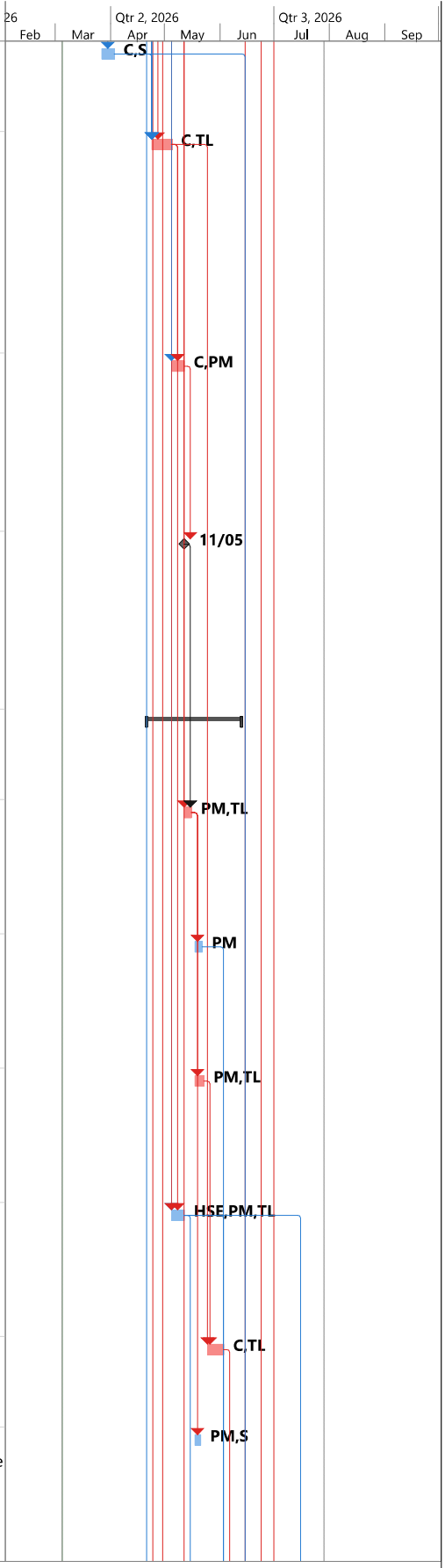
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15	2.6		Develop integrated block-flow diagram & system architecture	5 days	Thu 09/04/26	Wed 15/04/26	11,12,	Technical Lead
16	2.7		Identify supporting sytems (utilities, safety, I&C, monitoring)	3 days	Thu 16/04/26	Mon 20/04/26	15	Data & Instrumentation Lead,HSE Lead,Technical Lead
17	2.8		Technical review of concept design	3 days	Tue 21/04/26	Thu 23/04/26	16,15	Commercial/Market Lead,HSE Lead,Project Manager,Technical
18	2.9		Milestone 2: Concept design chosen	0 days	Thu 23/04/26	Thu 23/04/26	17	
19	**3**		**Market, stakeholder & commercial work**	**41 days**	**Mon 16/03/26**	**Mon 11/05/26**		
20	3.1		Market analysis for industrial H2 fuel in region	7 days	Mon 16/03/26	Tue 24/03/26	3	Commercial/Market Lead
21	3.2		Policy & regulatory review (UK ETS, H2 stratergy, pipelines)	5 days	Mon 16/03/26	Fri 20/03/26	3	Commercial/Market Lead,Regulatory/Peri Lead
22	3.3		Identify candidate Industrial heat users	3 days	Wed 25/03/26	Fri 27/03/26	20	Commercial/Market Lead,Stakeholder & Comms Lead
23	3.4		Initial engagement with target industrial user	5 days	Mon 30/03/26	Fri 03/04/26	22	Commercial/Market Lead,Stakeholder & Comms Lead
24	3.5		Develop outline H2 offtake concept (volumes, pricing, term)	6 days	Mon 06/04/26	Mon 13/04/26	23	Commercial/Market Lead
25	3.6		Feedstock supply mapping (waste oil/pyrolysi providers)	5 days	Fri 20/03/26	Thu 26/03/26	10	Commercial/Market Lead,Technical Lead



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Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Critical	
Project Summary	Manual Summary	Critical Split	
Inactive Task	Start-only	Progress	
Inactive Milestone	Finish-only	Manual Progress	

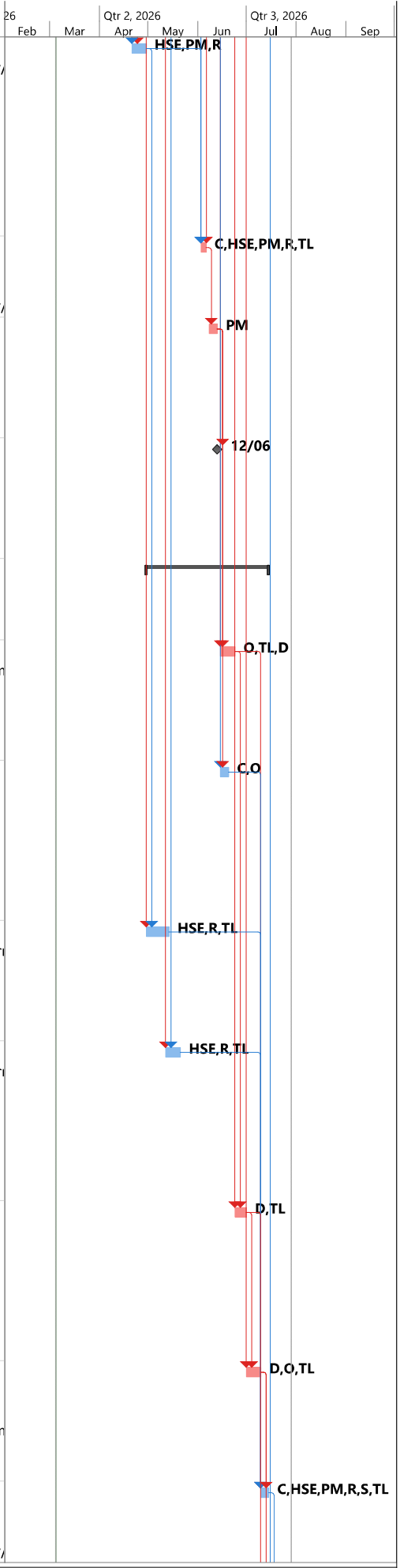
ID	WBS	Task Mode	Task Name	Duration	Start	Finish	Predece	Resource Names
26	3.7		Initial feedstock supplier engagement	5 days	Fri 27/03/26	Thu 02/04/26	25	Commercial/Market Lead,Stakeholder & Comms Lead
27	3.8		High level techno-economic assessment (H2 revenue, CO2 value, CAPEX/OPEX ranges)	7 days	Fri 24/04/26	Mon 04/05/26	26,24,2	Commercial/Market Lead,Technical Lead
28	3.9		Draft commercial/partr model (roles, responsibilities)	5 days	Tue 05/05/26	Mon 11/05/26	27,25	Commercial/Market Lead,Project Manager
29	3.10		Milestone 3: Commercial & stakeholder concept agreed in priciple	0 days	Mon 11/05/26	Mon 11/05/26	28	
30	**4**		**Detailed planning & approvals**	**39 days**	**Tue 21/04/26**	**Fri 12/06/26**		
31	4.1		Develop work breakdown structure (WBS)	4 days	Tue 12/05/26	Fri 15/05/26	29,17	Project Manager,Technical Lead
32	4.2		Develop detailed project schedule (Gantt)	4 days	Mon 18/05/26	Thu 21/05/26	31	Project Manager
33	4.3		Resource planning (people, equipment, external support)	5 days	Mon 18/05/26	Fri 22/05/26	31	Project Manager,Technical Lead
34	4.4		Refine risk assessment & mitigation plan	5 days	Tue 05/05/26	Mon 11/05/26	27,17,(	HSE Lead,Project Manager,Technical Lead
35	4.5		Cost estimate and budget proile	7 days	Mon 25/05/26	Tue 02/06/26	33,27	Commercial/Market Lead,Technical Lead
36	4.6		Define governance and reporting structure	3 days	Mon 18/05/26	Wed 20/05/26	31	Project Manager,Stakeholde & Comms Lead



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Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Critical	
Project Summary	Manual Summary	Critical Split	
Inactive Task	Start-only	Progress	
Inactive Milestone	Finish-only	Manual Progress	

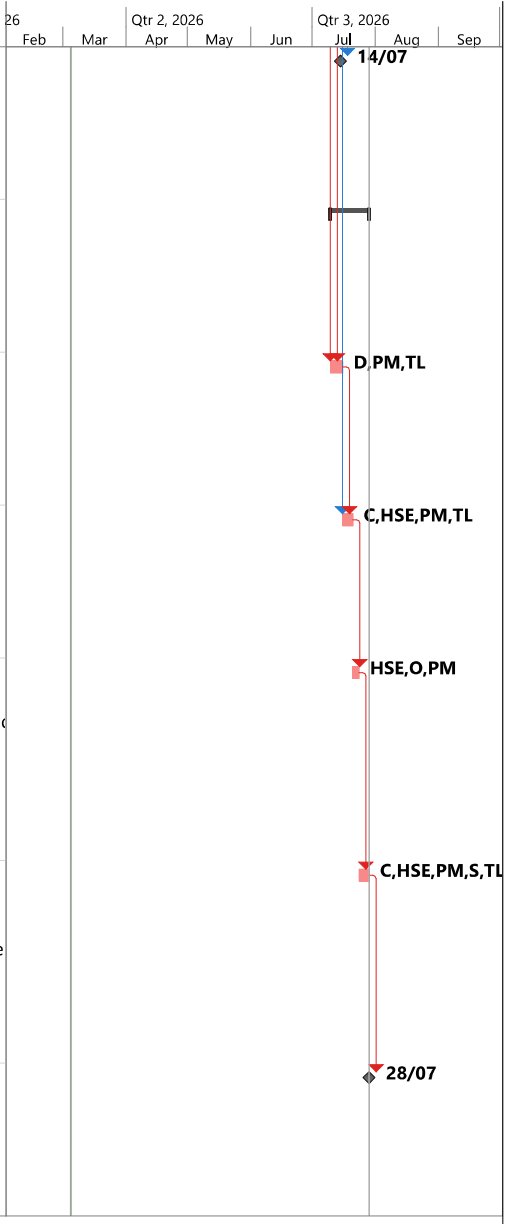
ID	WBS	Task Mode	Task Name	Duration	Start	Finish	Predece	Resource Names
37	4.7		Prepare regulatory and permitting stratergy (safety, environment, pipeline)	7 days	Tue 21/04/26	Wed 29/04/26	21,16	HSE Lead,Project Manager,Regulatory/ Lead
38	4.8		Internal review of plan and budget	3 days	Wed 03/06/26	Fri 05/06/26	37,35,:	Commercial/Market Lead,HSE Lead,Project Manager,Regulatory/
39	4.9		Client/sponsor approval of project plan	5 days	Mon 08/06/26	Fri 12/06/26	38	Project Manager
40	4.10		Milestone 4: Project plan approved / Launch ready	0 days	Fri 12/06/26	Fri 12/06/26	39	
41	**5**		**Implementation and pilot preperation**	**54 days**	**Thu 30/04/26**	**Tue 14/07/26**		
42	5.1		Site integration planning (layout, tie-ins, interfaces)	7 days	Mon 15/06/26	Tue 23/06/26	39,15	Operations Representitive,Techn Lead,Data & Instrumentation Lead
43	5.2		Feedstock logistics plan (contracts, tanker schedules, storage)	5 days	Mon 15/06/26	Fri 19/06/26	39,26	Commercial/Market Lead,Operations Representitive
44	5.3		H2 pipeline pre FEED study (route, hydraulics, safety)	10 days	Thu 30/04/26	Wed 13/05/26	37,13	HSE Lead,Regulatory/Peri Lead,Technical Lead
45	5.4		HSE management plan (hazards, controls, emergency response)	7 days	Tue 12/05/26	Wed 20/05/26	34,16	HSE Lead,Regulatory/Peri Lead,Technical Lead
46	5.5		Data and instrumentation plan (what to measure, when and how)	5 days	Wed 24/06/26	Tue 30/06/26	42,16	Data & Instrumentation Lead,Technical Lead
47	5.6		Draft pilot test plan (objectives, KPIs, run matrix)	7 days	Wed 01/07/26	Thu 09/07/26	46,17	Data & Instrumentation Lead,Operations Representitive,Techn Lead
48	5.7		Implementation and readiness review	3 days	Fri 10/07/26	Tue 14/07/26	47,45,4	Commercial/Market Lead,HSE Lead,Project Manager,Regulatory/



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Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Critical	
Project Summary	Manual Summary	Critical Split	
Inactive Task	Start-only	Progress	
Inactive Milestone	Finish-only	Manual Progress	

ID	WBS	Task Mode	Task Name	Duration	Start	Finish	Predece	Resource Names
49	5.8		Milestone 5: Ready for pilot implementation	0 days	Tue 14/07/26	Tue 14/07/26	48	
50	**6**		**Monitoring, evaluation and close out**	**13 days**	**Fri 10/07/26**	**Tue 28/07/26**		
51	6.1		Define monitoring and reporting framework	4 days	Fri 10/07/26	Wed 15/07/26	47,46	Data & Instrumentation Lead,Project Manager,Technical Lead
52	6.2		Define success criteria and evaluation plan	3 days	Thu 16/07/26	Mon 20/07/26	51,34	Commercial/Market Lead,HSE Lead,Project Manager,Technical Lead
53	6.3		Lessons learnt and continuous improvement process	3 days	Tue 21/07/26	Thu 23/07/26	52	HSE Lead,Operations Representitive,Projec Manager
54	6.4		Final project review and handover to implementation team	3 days	Fri 24/07/26	Tue 28/07/26	53	Commercial/Market Lead,HSE Lead,Project Manager,Stakeholde & Comms Lead,Technical Lead
55	6.5		Milestone 6: Project definition and plannnig complete	0 days	Tue 28/07/26	Tue 28/07/26	54	



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Task	Inactive Summary	External Tasks	
Split	Manual Task	External Milestone	
Milestone	Duration-only	Deadline	
Summary	Manual Summary Rollup	Critical	
Project Summary	Manual Summary	Critical Split	
Inactive Task	Start-only	Progress	
Inactive Milestone	Finish-only	Manual Progress	

PM Risk Assessment Framework

		PM					PM	PM				PM
Risk ID	Risk	Project Stage likely to occur	Probability of occurance	Impact of occurance	PI Score	Mitigation Actions	Include as Task?	Live When?	New Probability	New Impact	New PI Score	Risk Retired?
1	Budget overrun may occur caused by inderestimated costs (e.g. pipeline CAPEX, engineering effort, permitting fees) or scope creep that would result in the projectexceeding approved funding and requiring re-scoping or additional funding.	late stages	M	MH	12	Conservative cost estimating with contingencies, scope control and regular budget reviews by Market lead and Project manager during tasks 4.5 and 4.8 will reduce probability and impact of budget overrun.	Yes	12/05/2026	ML	M	6	No
2	Insufficient waste oil/pyrolysis feedstock may occur caused by tower than expected regional volumes or competing demand that would result in inability to run the HyPER pilot and failure to meet hydrogen production targets.	middle stages	M	H	15	Detailed feedstock supply mapping and engagement with multiple waste oil/pyrolysis suppliers by Market lead and technical Lead during tasks 3.6 and 3.7 will reduce probability of feedstock shortages.	Yes	20 March 2026	ML	MH	8	No
3	Insufficient or unstable hydrogen demand from target industrial heat user may occur caused by changes in production levels, energy prices or internal priorities that would result in under use of the HyPER pilot and poor economic performance.	late stages	M	H	15	Identifying multiple industrial heat users, engaement and development of a stable H2 offtake concept by Market lead and Stakeholder Lead during tasks 3.3,3.4 and 3.5 will reduce probability and impact of lower demand.	Yes	25 March 2026	ML	MH	8	No
4	Significant delays in obtaining regulatory approvals for the hydrogen pipeline may occur caused by complex permitting requirements, safety case approvals or stakeholer objections that would result in delayed implementation to commison the hydrogen delivery	middle stages	MH	MH	16	Development of dedicated regulatory and permitting stratergy, early engagement with regulators and intergration of requirements into preFEED by Regulatory lead,HSE lead and Project manager during tasks 4.7 and 5.3 will reduce probability and impact of the delays	Yes	21 April 2026	M	MH	12	No
5	A safety incident, such as hydrogen leak, explosion or feestock spill, may occur caused by inadequate design of safety systems, procedures or training that would result in potential injury, regulatory shutdown or reputation damage.	late stages	ML	H	10	HSE management plan, including hazard identification, safety systems and emergency procedures developed by HSE leas, technical lead and operations rep during tasks 2.7, 4.4 and 5.4 will reduce probability and impact of safety incidents.	Yes	16 April 2026	L	MH	4	No
6	Actual techno-economic performance may be worse than expected caused by higher CAPEX/OPEX, lower efficiency or lower capture rates than phase 1 assumptions that would result in uncompetitve hydrogen cost and a weak business case.	middle stages	M	MH	12	High level techno-economic assessment with sensitivity analysis on key cost an performance assumptions by Technical lead and Commercial lead during tasks 3.8 and 4.5 will reduce impact by clarifying feasibility and focussing on key high value design improvements.	Yes	24 April 2026	M	M	9	No
7	Key planning activities may overrun occuring because of inadequte resourcing, optimistic estimates of unforseen complexity that would result in delayed completion of the project definition or implementation and missed funding.	late stages	M	M	9	Development of realistic WBS, detailed schedule and resource plan with clear critical path and review periods by Project manager and Technical lead during tasks 4.1, 4.2 and 4.3 will reduce probability of schedule overrun.	Yes	12 May 2026	ML	M	6	No
8	Misalignment between Cranfield, the industrial heat user, feedstock suppliers and regulators may occur caused by unclear roles, objectives or communication that could result in scope changes, delays or loss of stakeholder support.	late stages	ML	MH	8	Establishing clear governance, reporting and stakeholder engagement plans and success criteria by Project manager and stake holder lead during tasks 4.6 and 5.6 will reduce probability of misalignment	Yes	18 May 2026	L	M	3	No
9	Inadequate data quality or monitoring scope may occur caused by insuffiecient instrumentation or poor data management that would result in inability to validate performance, quantify carbon benefits or support scale up decisions.	late stages	M	M	9	A detailed data instrumentation plan including defined KPIs, sensor requirements and data handling processes , by Data and instruumentation lead and Technical lead during tasks 5.5,5.7 and 6.1 will reduce impact of poor data quality	Yes	30 April 2026	ML	M	6	No
10	Community or local stakeholder concerns may occur caused by percived safety, traffic or environmental impacts that would result in additional scrutiny, delays or reputational damage.	late stages	ML	M	6	Development of a stakeholder engagement plan, including clear communication benefits and safety measures by Stakeholder lead and Project manager during task 5.6 will reduce probability and impact of community concerns.	Yes	30 April 2026	L	ML	2	No
11	Intergration complexity at the Cranfield site may occur caused by unanticipated tie-in constraints, space limitations or clashes with existing infrastructure that would result in additional design work, cost or schedule impact.	late stages	M	M	9	Early site integration planning, including layout studies and interface reviews by Technical Lead, Operations rep and Data and instrumentation lead during task 5.1 will reduce probability and impact of integration issues.	Yes	15 June 2026	ML	M	6	No
12	Key project learnings may not be captured or reused due to inadequate close out, documentation or reflection that would result in repeated mistakes and reduced value for future HyPER or industrial cluster projects.	late stages	ML	ML	4	Lessons learned workshop, evaluation plan and documentation handover by Project manager, HSE lead and Operations rep during tasks 6.2,6.3 and 6.4 will reduce probability and impact.	Yes	16 July 2026	L	ML	2	No

Individual Reflection:

During this module, I successfully created a complete project plan for the HyPER oil-pyrolysis hydrogen feasibility study, including a charter, design slide, comprehensive justification (with appendices), Gantt chart (45 tasks) and risk assessment (11 risks covering different project stages). Working in a group of 5, my role primarily focussed on project planning and market research, where I created the Gannt chart with predecessors and critical path and researched the hydrogen market. The group however, worked largely as a team throughout the project's tasks, from identifying key stakeholders to creating the risk assessment.

Development of key skills:

Teamwork: I actively contributed to group brainstorming sessions, particularly in the market analysis and risk definitions, where my team and I synthesised different ideas on commercial and HSE aspects which built a comprehensive risk assessment. A highlight was collaboratively refining the project plan during Day 3 to allow multiple tasks to run in parallel with balanced resource allocation, doing this process as a team strengthened our final product and mirrored real engineering team dynamics. A potential area for improvement however, is I tended to research any HyPER related queries after the sessions, whereas to be more helpful to the team, it would likely have been more helpful to ask questions in the session.

Communication: I believe presenting the Gantt chart logic after creation in MS Project to the group honed my ability to explain the more complex features simply (e.g. predecessors, WBS, resource allocation, critical path, etc.).

Problem solving: I decided to prioritise optimising the critical path by overlapping design tasks in the Gantt (e.g. feedstock mapping and HyPER interface definition), resulting in a shorter project duration and derisking through better alignment with the time constraint.

Time management: The group work was delivered on time with each member sticking to agreed deadlines. The individual assignment however, was poorly scheduled. I severely underestimated the time the justification section would take leading to working overtime close to the deadline, in the future I should plan more thoroughly, breaking down each part into subsections to obtain a better understanding of the potential project timeline.

Overall, I believe this module equipped me with the basic tools for not only project leadership roles but also in my own projects where time management and planning are crucial.

Appendix A: Feedstock Market and Alternative Analysis

A.1 Oil/Pyrolysis Feedstock Availability in UK

Waste oil and pyrolysis oil sources:

The UK generates substantial volumes of waste oils and pyrolysis oils suitable as HyPER feedstock, including:

- Waste cooking oil (UCO): Estimated ~200,000-250,000 tonnes/yr collected for biodiesel and other valorisation pathways (NNFCC, 2024 estimates)
- Industrial waste oils: Lubricants, hydraulic oils, transformer oils post-treatment.
- Pyrolysis oils from plastic/tyre waste: Emerging circular economy pathway, UK plastic wate to oil capacity expanding rapidly (e.g., recycling technologies, plastic energy projects).

Regional availability:

Region	Waste Oil Sources	Estimated Annual Volume (tonnes/yr)	Notes
East of England (Cranfield Region)	Food processing, light manufacturing, agricultural machinery	15,000-25,000	Proximity to Cranfield, road tanker delivery feasible within 50 km radius
Industrial Clusters (Teesside, Humber, Merseyside)	Petrochemical, refinery, manufacturing waste streams	50,000-100,000	Larger volumes, suited to commercial scale deployment (50-300 MWth)
Southeast (London, Thames Valley)	Commercial food service, logistics, manufacturing	30,000-50,000	High collection infrastructure but competing biodiesel demand

For pilot scale (700 kg H_2/day at 1.5 MWth HyPER):

- Estimated feedstock requirement ~2-3 tonnes oil/pyrolysis feedstock per day (depends on feedstock carbon content and HyPER efficiency).
- Annual feedstock: ~650-1000 tonnes/year (at 88% utilisation)
- Conclusion Regional waste oil availability in East of England sufficient to supply pilot for multiyear demonstration without competing with existing biodiesel pathways.

The regional feedstock bands reflect scaled estimates based on UK level waste and oil availability (~200,000-250,000 t/yr of UCO reported in industry sources [1]) and known industrial clustering rather than precise measurements. The ~2-3 t/day feedstock requirement is also an approximation based on pilot capacity and the expected outputs, exact value would be refined in detailed design.

A.2 Feedstock Comparison: Oil/Pyrolysis vs Alternatives

Feedstock Type	Oil/Pyrolysis (selected)	Biomethane	Natural Gas (fossil)
Carbon Intensity	Low (waste valorisation), CO_2 from fossil carbon fraction but circular economy benefit	Very low (renewable biogenic CO_2)	High
UK availability	Good in industrial regions, waste streams abundant, (200k+ tons/yr UCO alone) [1]	Limited biogas grid injection capacity ~6.5 TWh/yr, competing demand (transport, heat)	Excellent (national gas grid)
Cost (£/tonne or £/MWh) [need ref]	Variable, raw waste oils ~£100-300/tonne scaled down estimate from processed biodiesel-grade UCO (£400-700) [2]	Higher than natural gas	Lowest (commodity pricing, established market)
Suitability for HyPER/CHG	Proven for SE-SMR with liquid hydrocarbon feedstocks; requires pre-treatment (desulphurisation, volatiles control)	Ideal (methane rich, low contaminants), direct substitute for natural gas in HyPER	Baseline feedstock for HyPER Phase 1, proven technology
Logistics	Road tanker (ADR-compliant), schedulable delivery, no pipeline infrastructure required	Gas grid connection if available, otherwise liquid biomethane on tanker	Gas grid connection (standard)
Strategic fit	Circular economy, regional waste utilisation, supports UK waste to value policy	Aligns with renewable gas targets but limited scalability	Fossil fuel lock in, requires CCUS for low carbon H_2

Why oil/pyrolysis selected:

- Regional availability: Sufficient waste oil/pyrolysis feedstock East of England to supply pilot without major infrastructure investment.
- Circular economy benefit: Valorises waste streams, displacing virgin fossil feedstock.
- Cost effective: Waste oils competitively priced vs biomethane.
- Scalable: National waste oil volumes support future 50-300 MWth commercial deployment in industrial clusters.
- Technology readiness: HyPER/CHG adaptable to liquid hydrocarbon feedstocks with standard pre treatment

Alternatives rejected:

- Biomethane: Limited regional availability, higher feedstock cost, competing demand for transport/heat applications.
- Natural Gas: Fossil fuel pathway, inconsistent with waste valorisation and circular economy objectives of this pilot pathway design

Appendix B: H_2 Fuel Market and Policy Drivers

B.1 UK Industrial Heat Sector Profile

Market size and emissions:

- UK industrial sector accounts for ~13% of total UK GHG emissions in 2024 [3]
- Industrial heat (process heating, steam generation, drying, etc.) represents ~2/3 of industrial energy demand [4]
- Key sectors for H_2 fuel substitution [5]:
    - Food and drink processing: ~8 Mt CO_2/yr, widespread steam/heat demand, clustered facilities
    - Chemicals and pharmaceuticals: ~ 19 Mt CO_2/yr, high temperature process heat
    - Building products (cement, glass, bricks): ~11 Mt CO_2e/yr, high temperature kilns/furnaces
    - Metals: ~15Mt CO_2e/yr continuous high temperature operations

Target end user profile for pilot:

- Medium sized industrial heat user (food processing, light manufacturing)
- Heat demand: 5-10 MW thermal (continuous baseload suitable for ~700 kg H_2/day supply)
- Location: Within 10-50 km of Cranfield
- Regulatory pressure: Subject to UK ETS, net zero commitments, increasing carbon costs driving decarbonisation investment

B.2 UK Hydrogen Policy and Carbon Pricing

UK Hydrogen Strategy:

- 10 GW low-carbon H_2 production capacity by 2030 [6]
- Business models: Hydrogen production business model (HPBM) and net zero hydrogen fund (NZHF) providing >£1Bn support for early commercial projects [5]
- Low Carbon Hydrogen Standard: Certification for H_2 with lifecycle emissions <30 g CO_2e/MJ (vs ~90 g CO_2e/MJ for unbated SMR) [7]

UK ETS carbon price trends:

The UK ETS civil penalty price for 2026 is £49.41/t CO_2, up 18% from 2025s £41.84 [8]. This provides a clear economic incentive for industrial users to adopt low carbon H_2 fuel, avoiding carbon compliance costs estimated at ~£100k/yr for this pilots end user (see Appendix D).

Appendix C: Transport and Logistic Analysis

C.1 Feedstock Transport: Oil/Pyrolysis Delivery

Selected mode: Road Tanker (ADR-compliant)

- Feedstock requirement: ~2-3 tonnes/day (~650-1,000 tonnes/year at 88% utilisation)
- Delivery frequency: 1-2 tanker loads per week (assuming 10-15 tonne loads)
- Distance: <50 km from regional waste oil collection/processing facilities to Cranfield
- Cost: £100-200/tonne delivered (includes collection, treatment, transport), estimated based on common UK bulk collection prices for waste oil of ~£150-1200 per job [9]

Why road tanker selected:

- No pipeline infrastructure: Waste oils not integrated into existing industrial pipeline networks
- Schedulable: Flexible delivery scheduling compatible with batch storage and pilot operational flexibility
- ADR compliant: Standard hazardous goods transport regulation (UK ADR 2025), waste oils classified as class 3 flammable liquids or class 9 depending on composition
- Cost effective: Short haul road transport competitive for pilot scale (~1,000 tonnes/yr)

Alternatives considered:

- Rail: Not cost effective at pilot scale, requires rail sliding infrastructure
- Pipeline: No existing waste oil pipeline, high CAPEX for pilot scale volumes

C.2 Hydrogen Delivery: Pipeline vs Tube Trailer

Pilot-scale delivery requirement: ~700 kg H_2/day from HyPER plant to industrial heat user

Option 1: Dedicated Pipeline

Parameter	Value	Notes
Distance	5-10 km (Cranfield to nearby industrial user)	Assumed short haul local distribution
Capacity	~30 kg H_2/hour continuous	Sized for pilot 700kg/day output
Pressure	20-30 bar (medium pressure)	Matches HyPER output pressure (~24 bar design)
CAPEX	£500k - 1.5M (scaled down from larger system) [10]	£50-150k/km for small diameter H_2 pipeline (2-4 inch), includes materials installation, safety systems, permits
OPEX	£10-20k/yr (1.25% of CAPEX [10])	Inspection, maintenance, monitoring
Levelised cost	£0.3-0.5/kg H_2 (over 20 year life, 10% discount)	Lowest per kg cost for sustained volumes

Advantages:

- Lowest per-kg delivery cost for continuous supply
- Direct connection, no handling/logistics
- Scales efficiently to commercial deployment (same pipeline can handle 50+ MWth output with pssure boosting)

Disadvantages:

- Upfront CAPEX barrier
- Requires wayleaves, permits, safety cases
- Single point to point connection (limited flexibility)

Option 2: Tube Trailers (Compressed H_2)

Parameter	Value	Notes
Capacity per trailer	~300-500 kg H_2 at 200-500 bar	Standard Type 1 tube trailer
Delivery frequency	1-2 deliveries per day	700 kg/day requires ~2 trailer loads
Distance	5-10 km	Same routine as pipeline option
CAPEX	£200-300k, estimated based on [11]	Tube trailer fleet (203 trailers for rotation), user-site decompression/buffer storage
OPEX	£150-250k/yr [11]	Driver costs, fuel, trailer maintenance, compression energy at HyPER site (to 200-500 bar)
Levelised cost	£1.00-£1.50/kg H_2	~3x higher than pipeline for short distance / sustained volume

Advantages:

- Lower upfront CAPEX
- Flexibility (can serve multiple users with trailer fleet)
- Standard logistics model

Disadvantages:

- Higher per kg cost for continuous supply
- Requires high pressure compression at HyPER site (additional energy penalty ~15% of H_2 output)
- Logistics complexity (diver scheduling, safety protocols, trailer availability)

C.3 Decision: Pipeline Selected for primary H_2 Delivery

Rationale:

For sustained , continuous supply of 700 kg/day to a single industrial heat user over a 5-10 km distance, dedicated pipeline has:

- ~50-70% lower levelised cost (£0.3-0.5/kg vs £1.00-1.50/kg)
- Operational simplicity: No daily logistics, driver scheduling or trailer management
- Scalability: Same pipeline infrastructure supports future scale up to 50+ MWth commercial plant (only needs higher input pressure)
- Demonstration value: Proves low cost H_2 delivery model for UK industrial clusters

Trade off accepted:

- Higher upfront CAPEX (£0.5-1.5M) justified by 20 year asset life and significantly lower operating costs
- Pilot project can leverage UK hydrogen infrastructure funding (NZHF, HPBM) to offset pipeline CAPEX

C.4 Transport Related CO_2 Emissions vs Plant Carbon Benefit

Purpose: assess whether transport emissions materially affect the overall carbon benefit of the pathway.

Feedstock Transport Emissions (Road Tanker)

Assumptions:

- Feedstock requirement: ~1,000 tonnes/yr
- Average delivery distance: 30 km (round trip: 60 km)
- Delivery frequency: ~50 deliveries/yr (20 tonne loads)
- DEFRA factor: Using DEFRA'a 'all HGVs, average larden' factor of ~0.89 kg CO_2e/km [12]

Calculation:

- Vehicle km per year: 50 trips * 60km/trip = 3,000 km/yr
- Annual transport CO_2: 3,000 * 0.89121 = 2,674 kg CO_2e/yr ~ 2.7 tonnes CO_2e/yr

H_2 Pipeline (selected option): Zero Transport Emissions

From Appendix D (where ~2,093 tCO_2/yr avoided is assumed) transport emissions are <1% of avoided CO_2 , so transport is negligible compared to the process benefit.

Appendix D: Project Economics

D.1 Purpose and Scope:

This appendix shows feasibility economic calculations for the proposed HyPER oil/pyrolysis to hydrogen pilot plant to show the commercial viability and support the justification in the slides. These are approximate estimates based on assumed pilot capacity and current UK market conditions.

D.2 Key Assumptions

Parameter	Value	Source/Justification
H_2 production capacity	700 kg H_2/day	Assumed pilot scale suitable for single industrial heat user, consistent with small scale demonstration plant sizing. [mod materials]
Plant capacity factor	88%	Consistent with HyPER Phase 1 economic analysis basis ~8,000 operating hours/yr [13]
Industrial low carbon H_2 fuel price (UK)	£4.00/kg	Blue hydrogen (SMR+CCS) estimated at £2.50-£4.50/kg, delivered low carbon H_2 for industrial applications typically £3.00-£5.00/kg including logistics. [14]
UK ETS carbon price	£49/t CO_2	UK ETS civil penalty carbon price for 2026 reporting year is £49.41/t CO_2 [8]
Natural gas CO_2 emissions factor	0.185 kg CO_2/kWh	Standard UK natural gas combustion emissions factor [15]
Hydrogen lower heating valve	120 MJ/kg (33.3 kWh/kg)	Standard property
CO_2 displacement factor	~9.3 kg CO_2 avoided per kg H_2 used for heat	Calculated: 1kg H_2=33.3 kWh thermal energy, displacing gas at 0.185 kg CO_2/kWh avoided ~6.2 kg CO_2/kg H_2 (combustion only). Extra ~3.1 kg CO_2/kg H_2 avoided from HyPER's 98% CO_2 capture vs conventional SMR emissions. [13]

D.3 Annual Hydrogen Production

Calculation:

Operating days per year: 700 kg/day * (8000 h/yr / 24 h/day) ≈ 700 kg/day * 333 days ≈ 230,000 kg H_2/year (225t H_2/yr taken as conservative estimate).

D.4 Hydrogen Revenue Calculation

Base case (£4.00/kg):

Annual H_2 revenue: 225,000 * £4.00 = £900,000/year, H_2 revenue ~£900k for a 700 kg/day pilot plant operating at ~88% utilization.

D.5 Carbon Displacement Value

Industrial heat user substitutes low carbon H_2 fuel for natural gas. Two carbon benefits:

1. End use displacement: Natural gas combustion benefits avoided (0.185 kg CO_2/kWh/kg H_2=6.2kg CO_2 per kg H_2)

2. Production emissions avoided: HyPER captures 98% of process CO_2 vs conventional SMR (~9 kg CO_2/kg H_2 unbated), net benefit ≈ 3.1 kg CO_2/kg H_2.

Total system displacement ~9.3 kg CO_2 avoided per kg H_2 consumed.

Calculation:

1. Annual H_2 production 225,000 kg/yr, CO_2 avoided 225,000 * 9.3 = 2,093 tonnes CO_2/yr.
2. UK ETS carbon price: £49/t CO_2, annual carbon value: 2,093 * 49 ≈ £103k/yr

CO_2 displacement value is ~£100k/yr, representing a ~10% increase on top of H_2 fuel revenue in the form of avoided carbon compliance costs for the industrial user.

This results in ~£1.0M/yr in combined fuel revenue and carbon value for the 700 kg H_2/day plant which shows commercial scale viability potential.

D.6 Competitive Position and Techno Economic Basis

The HyPER /CHG technology underpinning this pathway offers significant techno-economic advantages vs conventional SMR+CSS [13]:

- ~56% lower CAPEX (£105M vs £240M for 200 MWth single train plant)
- ~20% lower LCOH (£170/kNm^3 vs £212/kNm^3 for high pressure case)
- Up to 98% CO^2 capture vs 90% for SMR+CCS
- Carbon footprint ~40% of counterfactual (38.1 vs 90+kg CO_2/kNm^3)

These advantages, validated in the HyPER phase 1 feasibility study support the long term cost competitiveness and scalability of the oil/pyro-HyPER pathway. The pilot scale demonstration de risks future commercial deployment at 50-300 MWth scale for industrial clusters.

D.7 Limitations and Next Steps

- These calculations are estimates for feasibility/design decision making.
- Full FEED study would refine:
    1. Detailed CAPEX, OPEX for pilot plant (feedstock, utilities, labor, maintenance)
    2. Contractual H_2 offtake price with industrial user
    3. Carbon accounting methodology and verification for UK ETS compliance
    4. Sensitivity to natural gas price, electricity costs and H_2 market dynamics
- Pilot operation would generate real world data to validate assumptions and support investment case for commercial scale deployment

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